Text Classification Proposal: From Bag of Words to Pre-trained Language Models

Section0: Team Members & Responsibilities

Group leader: 2213633- Zhai yukun || be responsible for all work.

Section 1: Background & Motivation

In the field of Natural Language Processing, text processing is arguably one of the most fundamental and important tasks. Throughout its development, it has undergone multiple paradigm shifts—from rule-based methods to traditional machine learning, then to deep learning, and further to pre-trained language models. This project aims to systematically review and introduce the entire technological evolution of text classification, compare the differences among various technical approaches, and provide analysis and evaluation.

Section 2: Problem Definition / Research Questions

The main research questions are as follows:

Question 1: What are the core ideas and representative methods at each stage of the development of text classification technology?

Question 2: How do different text representation methods perform in terms of performance and computational cost on classification tasks?

Question 3: At the current stage, what difficulties and possibilities remain in the development of text processing technology?

Section 3: Expected Outcomes

Through the research, organization, and analysis of this project, we hope to clarify the entire development history of text processing technology, understand the characteristics, advantages, and limitations of each stage, and on this basis, further reflect on the possible future development paths and technological prospects of text processing.

Section 4: Project Timeline

Stage 1: Review relevant literature in the field and understand the technical characteristics of each stage of text processing.

Stage 2: Organize and compare the characteristics of technologies at each stage, including their advantages, problems, and shortcomings.

Stage 3: On this basis, reflect on the current development situation and challenges of text processing technology, and consider what possibilities may exist for the future of this field.